

Video Assignment

An application of linear algebra

2201NSC Linear Algebra and Applications

The assignment

Produce a **10-minute video** exploring a real-world application of linear algebra.

Primary goal

Use ideas from this course to investigate how linear algebra helps solve a real-world problem and then explain the application to someone else in the course.

The video should **extend** the concepts and content covered in class, rather than repeat an example or explanation that we have already completed.

Groups

Work in a group of **two**.

- ▶ A group of three may be permitted where necessary to ensure that everyone can join a group.
- ▶ Form your group by the **end of class in Week 9**.
- ▶ Anyone without a group at that time will be assigned to one.

Depth and progression

The video should develop the topic in a clear sequence.

Motivation

Introduce the application and explain the problem that linear algebra helps to solve.

Simple example

Use a manageable example to explain the mathematical process clearly.

Extensions

Move to more advanced examples and investigate what changes.

The advanced material should build on the simple example, rather than become an unrelated second topic.

Sharing the presentation

The assignment is a shared project. Each group member should contribute roughly equally to the work as a whole.

- ▶ Share the planning, mathematical investigation, preparation of visuals, and production of the final video.
- ▶ Each person should appear for roughly the same amount of time.
- ▶ The person speaking must be **visible** in the video.
- ▶ Each speaker should explain part of the mathematics or its application.

Different tasks do not need to be divided in exactly the same way, but everyone should make a meaningful and comparable contribution.

Visual communication

Each group should create slides or use a similar visual aid.

The visuals should help the audience follow:

- ▶ the practical problem;
- ▶ the important matrices, vectors, calculations, or diagrams;
- ▶ the results of the investigation.

Keep mathematical expressions readable and leave each visual on screen long enough for the audience to understand it.

Sample topic: GPS positioning

The topic

Data from transmitters can be used to estimate a person's position.

Starting point

Explain how transmitter data can be used to calculate a position in two dimensions. Use a simple example with a known position to demonstrate that the method works.

Further investigation

Add measurement error and examine its effect on the calculated position. Compare transmitter arrangements and investigate which arrangements produce more reliable estimates.

Sample topic: Recommending a song

The topic

A matrix of likes, dislikes, or ratings can be used to recommend a song to a person.

Starting point

Construct rating vectors or a rating matrix and use them to make a recommendation.
Use a simple example to demonstrate that the method produces a reasonable prediction.

Further investigation

Explore how to represent disliked and unrated songs. Test an extension such as centred ratings, a selected number of nearest neighbours, or similarities between songs. Identify cases where the method works well and where it makes a poor recommendation.

Sample topic: Errors in transmitted messages

The topic

The null space of a matrix helps detect and correct errors introduced during the transmission of a message.

Starting point

Choose a suitable encoded message from the null space of a check matrix. Explain how a recipient checks the message and show how the calculation confirms that it contains no detectable errors.

Further investigation

Show how the method detects and corrects a transmission error. Then extend the example to a longer message or investigate what happens when two bits are corrupted, including a case that cannot be corrected reliably.

Alternative topic: Your choice of application

The topic

Your group may propose another real-world application of linear algebra that you would like to investigate.

Starting point

Identify the practical problem and the linear algebra needed to investigate it. The topic should extend material from the course and support both a simple explanation and a more advanced investigation.

Approval required

Discuss the proposed topic with the course convenor and receive approval before beginning the investigation.

What matters in the video

The focus is the **content and application**.

Production quality

You do not need professional recording equipment, elaborate editing, special effects, or expensive software.

Your mark will reflect the quality of the content and application rather than the polish of the recording. The audience should still be able to see the speaker, read the visual material, and hear the explanation clearly.